Existing System Comparison

Khan Academy with Khanmigo, Prodigy Math, IXL and Bilimland are compared against nine consistent criteria. The matrix informs MathVerity's positioning and identifies the limits of claims about comparable products.

Four relevant products were selected: Khan Academy with Khanmigo, Prodigy Math, IXL and Bilimland, a platform from Kazakhstan, the proposed pilot context. This exceeds the minimum number of independent systems. The comparison concerns publicly described capabilities as of 22–23 October 2026. Other local platforms were not included because their features could not be confirmed from the public pages available to the team. Restricted dashboards and paid modes were not tested directly; pricing, language and regional conditions must be checked before a pilot. "Not confirmed" means that the reviewed sources provide insufficient evidence, rather than that a feature is absent.

Comparison Against Nine Criteria

Table 1. Comparison of existing systems

Criterion	Khan Academy and Khanmigo [P01–P03]	Prodigy Math [P04–P06]	IXL [P07–P09]	Bilimland [P10]	MathVerity (planned)
Audience and mathematics	● Math content; AI help on material	● Math, grades 1–8	● Math skills and diagnosis	● Math grades 1–6, curriculum-based	◐ Grades 5–6; 4 blocks, 8 skills
Personalisation	● Skill mastery; help on current material	● Adaptive practice in a game	● Diagnosis and recommendations	○ Adaptive practice not confirmed	◐ Explicit skill statuses; ML later
Feedback	● AI guides the solution; explanations	◐ In-game questions; AI chat not confirmed	◐ Practice, analytics; AI chat not confirmed	◐ Tests and results; AI help not confirmed	● Restricted AI hints with static fallback
Gamification	◐ Mastery points and badges	● Game world and rewards	◐ Awards	◐ Mentioned generally; details not confirmed	◐ XP and achievements
Parent access	● Child accounts, progress, Khanmigo history	● Parent tools and reports	◐ Home analytics	○ Not confirmed	● Report with evidence for each status
Skill diagnosis	◐ Mastery levels from practice and tests	◐ Adaptive items; progress information	● Diagnostic and practice plans	○ Not confirmed	◐ Statuses from stored evidence

Criterion	Khan Academy and Khanmigo [P01–P03]	Prodigy Math [P04–P06]	IXL [P07–P09]	Bilimland [P10]	MathVerity (planned)
Independent transfer after assistance	○ Matching protocol not confirmed	○ Not confirmed	○ Not confirmed	○ Not confirmed	● Independent check after help
Access to core features	◐ Library free; Khanmigo separate	◐ Content free; some parent tools by membership	○ Subscription or school licence	◐ Purchase option; terms not stated	Not yet defined; pilot use only
Local language and curriculum	○ Not confirmed	○ Not confirmed	○ Not confirmed	● Kazakh, Russian, English; curriculum-based	◐ Languages planned; curriculum alignment not claimed

Key: ● confirmed in the reviewed sources; ◐ partly confirmed or limited; ○ not confirmed (insufficient evidence, not proof of absence). The MathVerity column shows planned MVP features that are not yet implemented, and its ratings are the team's own assessment.

Design Implications

Khanmigo is a close comparator for pedagogically guided AI assistance and parent access. A combination of AI and a parent dashboard therefore cannot be claimed to be absent from the market. Prodigy illustrates the close integration of practice with a game loop; IXL illustrates detailed diagnosis and recommendations. This matrix does not establish MathVerity's superiority over these systems.

Bilimland is the closest local reference. It states curriculum-based mathematics lessons for grades 1–6 in Kazakh, Russian and English, but the reviewed page does not confirm adaptive practice, parent access or diagnostic reports [P10]. This supports the hypothesis of a local gap, which still requires confirmation through interviews.

The proposed positioning is a compact Kazakh- and Russian-language flow for selected skills with an explicit distinction between assisted and independent work, structured error diagnosis and a report that explains its evidence. Its usefulness to local families remains a hypothesis. Alignment with the national curriculum is not claimed until it has been checked.

The experimental baseline is an otherwise identical practice tool developed within the project, using static hints. This keeps content, time and interface

constant. External products inform market analysis and UX design but are not used as an uncontrolled experimental group.

References

[P01] Khanmigo for learners. https://www.khanmigo.ai/learners

[P02] Khanmigo for parents. https://www.khanmigo.ai/parents

[P03] Khan Academy. What is self-paced Mastery. https://support.khanacademy.org/hc/en-us/articles/360007253831-What-is-self-paced-Mastery ; Badges: https://www.khanacademy.org/badges

[P04] Prodigy Math. https://www.prodigygame.com/main-en/prodigy-math

[P05] Prodigy for parents. https://www.prodigygame.com/main-en/parents

[P06] The Parents' Guide to Prodigy Math. https://www.prodigygame.com/main-en/blog/parents-guide-to-prodigy

[P07] IXL. Analytics. https://ca.ixl.com/membership/teachers/analytics

[P08] IXL. Using IXL Analytics at Home. https://blog.ixl.com/2018/09/21/using-ixl-analytics-at-home/

[P09] IXL awards. https://www.ixl.com/awards

[P10] Bilimland. Educational platform for pupils and students (interactive lessons, e-textbooks, tests). https://bilimland.kz